

Module 12 - chapter 22

Complete CI/CD with Terraform - Part 1

The project files for the next three lectures can be found here:

Starting code:

<https://gitlab.com/twn-devops-bootcamp/latest/12-terraform/java-maven-app/-/tree/feature/starting-code>

Final project:

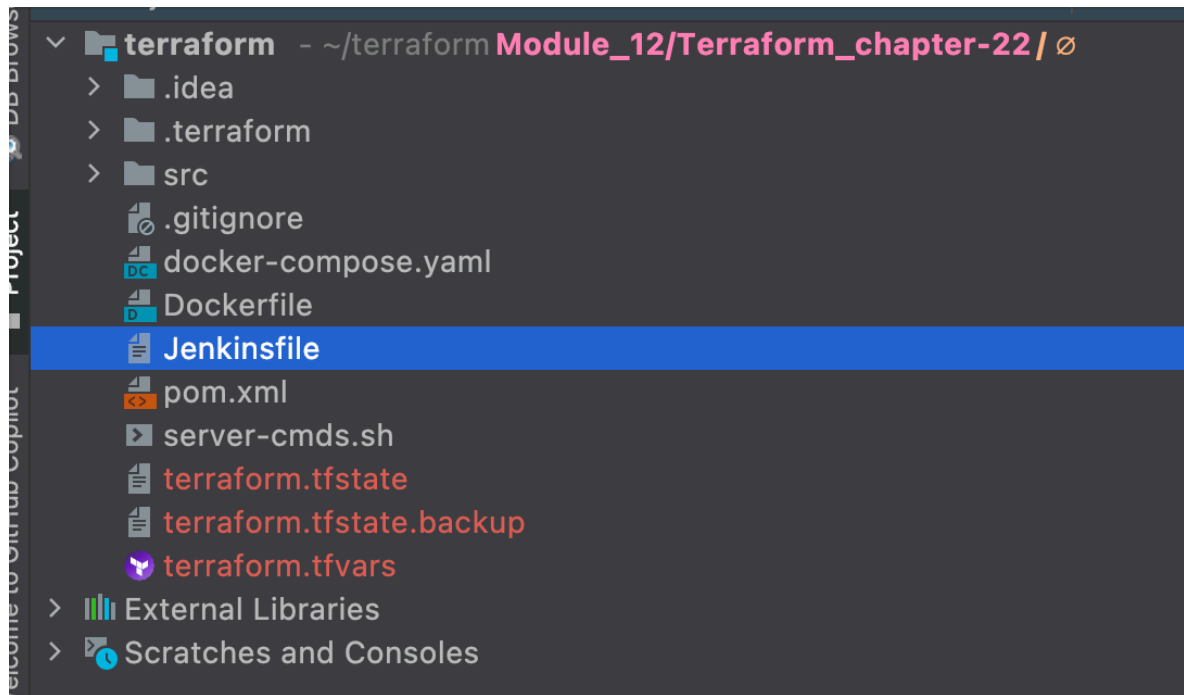
<https://gitlab.com/twn-devops-bootcamp/latest/12-terraform/java-maven-app/-/tree/jenkinsfile-sshagent>

I added the starting-code gitlab repo as a branch in my GitHub branch to keep all the chapters in the same repo

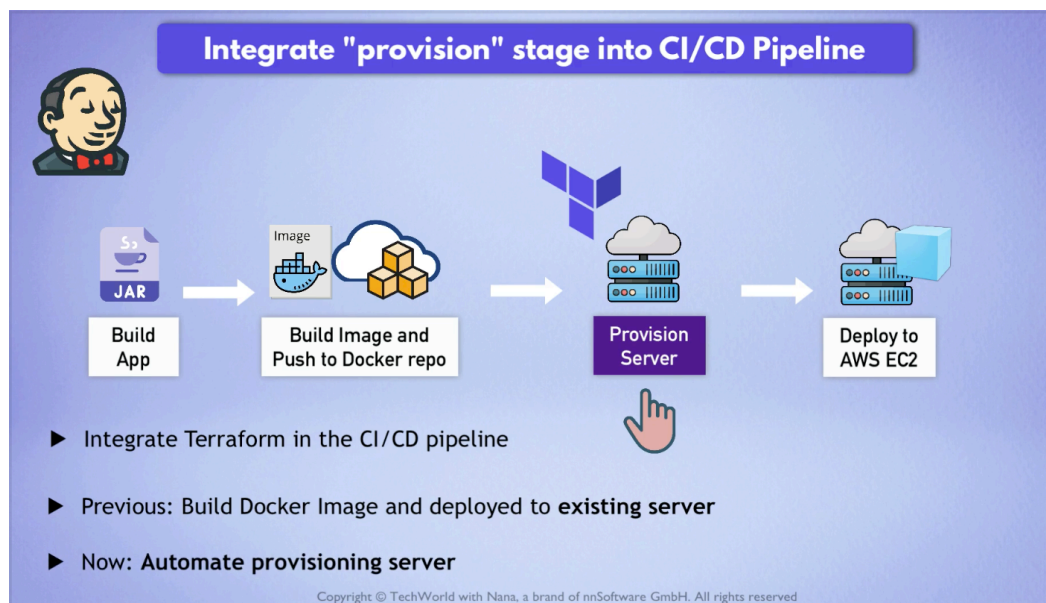
I added GitLab as a temporary remote and fetched just that branch:

```
git remote add gitlab <gitlab-repo-url>
git fetch gitlab <branch-name>
git checkout -b <branch-name> gitlab/<branch-name>
git push origin <branch-name>
git remote remove gitlab
```

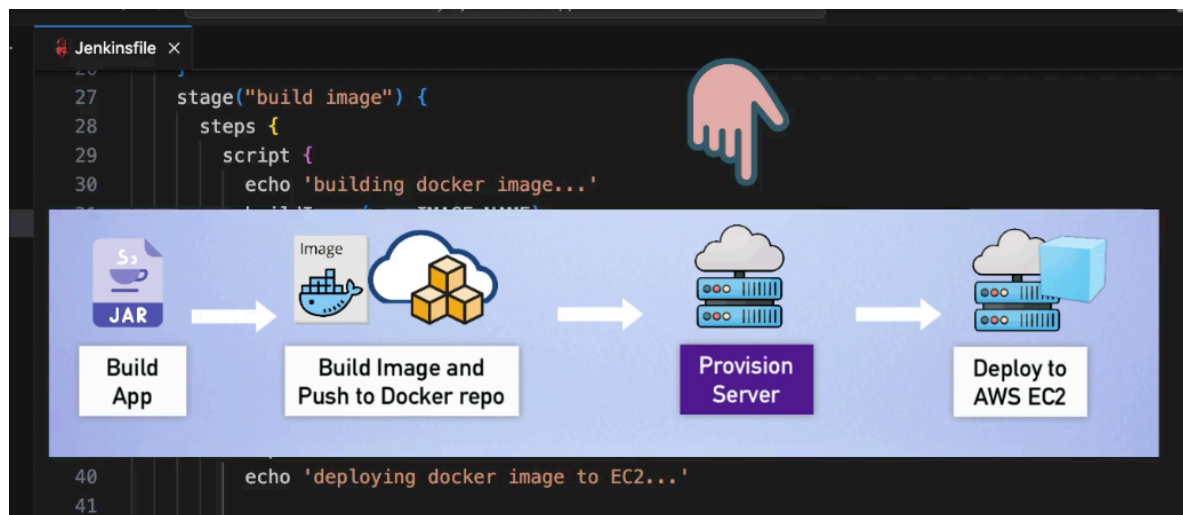
```
-> git remote add gitlab https://gitlab.com/twn-devops-bootcamp/latest/12-
terraform/java-maven-app.git
-> git fetch gitlab feature/starting-code
-> git checkout -b Module_12/Terraform_chapter-22 gitlab/feature/starting-
code
-> git push origin Module_12/Terraform_chapter-22
-> git remote remove gitlab
```



DEMO Overview



We will provision a server using terraform in an automated way



We will add a stage to the jenkinsfile called 'provision server' so the creation of the remote server is triggered by jenkins

```
35 }
36 }
37 stage("provision server") {
38   // tf provision server
39 }
40 stage("deploy") {
41   steps {
```

We will need to create an ssh key pair for the server so then we can ssh into that server.

Key pair per Ec2 instance.

Install tf inside jenkins container, so we can execute terraform commands inside the jenkins container

```
37 stage("provision server") {
38   // tf provision server
39   sh "terraform init"
40 }
```

Then we will create tf config files inside our java project, so we can execute terraform apply on files inside a folder in my project.

So tf config files will be part of our application code -> best practice (include everything the app needs)

Then we can adjust the jenkinsfile

Jenkinsfile

```
27 stage("build image") {
28   steps {
29     script {
30       echo 'building docker image...'
31       buildImage(env.IMAGE_NAME)
32       dockerLogin()
33       dockerPush(env.IMAGE_NAME)
34     }
35   }
36 }
37 stage("provision server") {
38   // tf provision server
39   sh "terraform init"
40 }
41 stage("deploy") {
42   steps {
43     script {
44       echo 'deploying docker image to EC2...'
45     }
46   }
47 }
```

Steps to do



- 1) Create SSH key-pair
- 2) Install TF inside Jenkins container
- 3) TF configuration to provision server
- 4) Adjust Jenkinsfile